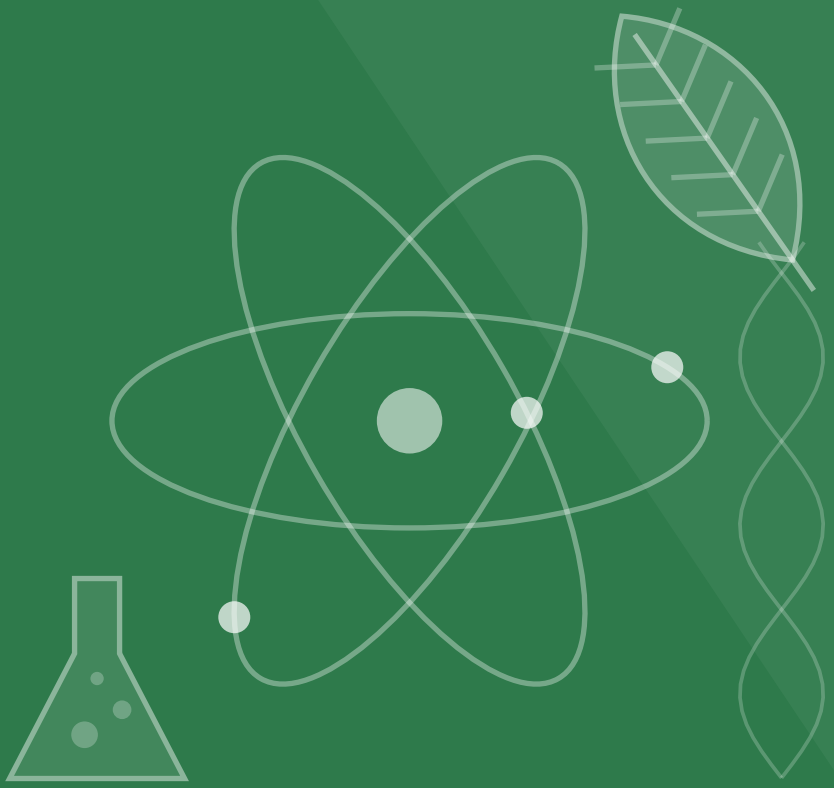
VENVLEARNING

STUDENT WORKBOOK



Science

GRADE 7

Daily practice, weekly checks, and unit tests

2 units

A VenvLearning family homeschool curriculum

GRADE 7 · SCIENCE

VENVLEARNING

Grade 7
Science

STUDENT WORKBOOK

Unit 1 Cycles of Matter and Energy

Unit 2 Properties and Changes of Matter

A VenvLearning family homeschool curriculum

How to use this workbook

Every lesson in your textbook has matching practice pages here. Show your thinking in the work space. When a problem has an answer line, write your final answer there.

- Warm-Up reviews something you already know.
- Practice uses today's lesson.
- Think Deeper problems ask you to explain or find a mistake.
- Exit Ticket shows your teacher what you learned today.

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UNIT 1

Cycles of Matter and Energy

Science practices, Earth's spheres, the water, carbon, oxygen, and nitrogen cycles, photosynthesis, respiration, food webs, and biodiversity.

Your practice pages for Unit 1. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

independent variable	The variable that is changed.
dependent variable	The variable that is measured.
photosynthesis	Light + CO_2 + water → glucose + oxygen.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
nitrogen fixation	Changing N_2 into usable forms.
biodiversity	The variety of life in an area.

LESSON 1.1 · PRACTICE
Week 1 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.1: Variables, Data, and Scientific Explanations

Name _____ Date _____

TODAY'S GOAL

I can identify independent, dependent, and controlled variables and write a scientific explanation using CER.

Standards: SEP 3, 4, 6

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.1 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Identify the Variables, Part E Analyze Data, Part F Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. The variable you change

2. The variable you measure

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · TEMPERATURE AND DISSOLVING

How does water temperature affect dissolving time?

You need: 3 identical effervescent antacid tablets (or 3 equal spoonfuls of sugar); 3 clear cups; cold, room-temperature, and warm tap water; thermometer; stopwatch.

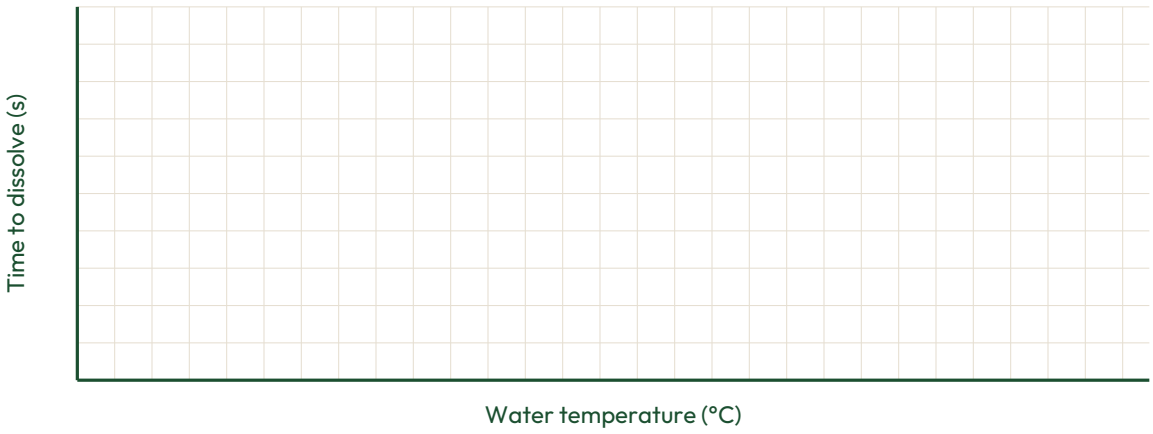
1. Measure 200 mL of water into each cup and record each temperature. ☐
2. Drop one tablet into each cup at the same moment and start the timer. ☐
3. Record the time when each tablet is completely dissolved. ☐
4. Repeat if you can, and average the trials. ☐

Safety: Use warm tap water, not boiling water. Do not swallow the tablets or solutions.

3 Record your results in the table.

Cup	Water temperature (°C)	Time to dissolve, trial 1 (s)	Trial 2 (s)	Average (s)

4 Graph your results.



5 What is the relationship between temperature and dissolving time?

PART C Check Your Understanding

Directions: Answer each question.

6 Plant height is measured after different amounts of fertilizer. Independent variable?

7 Same experiment: dependent variable?

8 Name two controlled variables.

PART D Identify the Variables

Directions: For the question, name the independent, dependent, and controlled variables.

9 Does the height of a ramp affect how far a car rolls? Independent:

10 Same: dependent:

11 Same: two controlled variables:

12 Write a hypothesis for the ramp experiment.

PART E Analyze Data

Directions: Use the data to answer the question. Describe the pattern and explain it.

13 Data: 0 g salt → water boils at 100.0°C; 20 g → 100.3°C; 40 g → 100.7°C. Write a CER.

PART F Exit Ticket

14 Why must controlled variables stay the same?

LESSON 1.2 · PRACTICE
Week 1 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.2: Earth's Spheres and the Water Cycle

Name _____ Date _____

TODAY'S GOAL

I can describe Earth's four spheres and model how energy from the sun and gravity drive the water cycle.

Standards: MS-ESS2-4

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.2 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the term that completes each line.

1. All living things make up the...

2. Water vapor cooling into droplets is...

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · WATER CYCLE IN A BAG

Can you model the water cycle?

You need: a zip-top bag; 1/4 cup water with a few drops of blue food coloring; tape; a sunny window; marker.

1. Draw a sun, cloud, and waves on the bag. ☐
2. Pour in the water and seal the bag. ☐
3. Tape it to a sunny window. ☐
4. Observe it after 1 hour, after 1 day, and after 2 days. ☐

3 Observations (Day 1 and Day 2):

4 Sketch the bag. Label where evaporation, condensation, and precipitation happened.

PART C Check Your Understanding

Directions: Answer each question.

5 Which sphere includes clouds and air?

6 What process returns water vapor from plants?

7 What force makes rain fall and rivers flow?

PART D Practice

8 Which sphere does a river belong to?

9 Which sphere does a mountain belong to?

10 What provides the energy for evaporation?

11 What happens to water that soaks into soil?

12 Describe one interaction between two spheres in the water cycle.

13 Explain the droplets on the inside of the bag using the words evaporation and condensation.

PART E Exit Ticket

14 Name the two things that drive the water cycle and what each does.

WEEKLY CHECK
After Lesson 1.2

About 15 minutes

Weekly Check 1: Variables and the Water Cycle

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Name each variable, or answer the question.

1. Does the amount of sunlight affect bean plant growth? Independent variable:

2. Dependent variable:

3. One controlled variable:

4. Water vapor cooling into droplets is...

5. Which sphere includes all living things?

6. Name the two drivers of the water cycle and what each does.

LESSON 1.3 · PRACTICE
Week 2 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.3: The Carbon and Oxygen Cycles

Name _____ Date _____

TODAY'S GOAL

I can explain how carbon and oxygen cycle among living things, the atmosphere, the oceans, and rocks.

Standards: MS-LS2-3

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.3 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the term for each description.

1. Coal, oil, and natural gas are...

2. Organisms that break down dead matter

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · FOLLOW A CARBON ATOM

What paths can one carbon atom take?

You need: station cards: Atmosphere, Plant, Animal, Decomposer, Ocean, Fossil Fuel; one die.

1. Start at the Atmosphere station. ☐
2. Roll the die; each station card tells where a roll sends you and why (e.g. 1–3: a plant takes you in during photosynthesis). ☐
3. Record each move for 10 rolls. ☐
4. Compare paths with a partner. ☐

3 Draw your carbon atom's path as a flow chart. Write the process that moved it on each arrow.

PART C Check Your Understanding

Directions: Answer each question.

4 Which process removes carbon dioxide from the air?

5 Name two processes that release carbon dioxide.

6 About what percent of air is oxygen?

PART D Practice

7 A tree takes in CO_2. Process?

8 A car burns gasoline. Process?

9 A fungus breaks down a log. Process?

10 You exhale. Process?

11 Explain why burning fossil fuels adds carbon to the atmosphere faster than natural processes remove it.

PART E Exit Ticket

12 How are the carbon and oxygen cycles connected?

LESSON 1.4 · PRACTICE
Week 2 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.4: The Nitrogen Cycle

Name _____ Date _____

TODAY'S GOAL

I can explain how nitrogen moves from the air into living things and back.

Standards: MS-LS2-3

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.4 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the term that completes each line.

1. Living things use nitrogen to build... _____

2. A bean plant is a... _____

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · ROOT NODULE SEARCH

Where do nitrogen-fixing bacteria live?

You need: a bean or pea plant (or photos from your teacher); magnifying glass; water.

1. Gently dig up the plant and rinse the soil from its roots. ☐
2. Look for small round bumps (nodules) on the roots. ☐
3. Sketch the roots and nodules. ☐

3 Sketch the roots and nodules you found. Label them.

4 What do the bacteria in the nodules do?

PART C Check Your Understanding

Directions: Answer each question.

5 What percent of air is nitrogen?

6 What does nitrogen fixation do?

7 Why do farmers plant legumes?

PART D Practice

8 Can animals use nitrogen gas directly?

9 How do animals get nitrogen?

10 Name a non-living way nitrogen is fixed.

11 What returns nitrogen from dead organisms to soil?

12 Explain why rotating crops with legumes can help soil.

PART E Exit Ticket

13 Trace one nitrogen atom from the air into your body.

WEEKLY CHECK
After Lesson 1.4

About 15 minutes

Weekly Check 2: Carbon, Oxygen, and Nitrogen

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

1. Which process removes CO_2 from air?

2. Name a process that releases CO_2.

3. About what percent of air is nitrogen?

4. What do nitrogen-fixing bacteria do?

5. Why does burning fossil fuels change the carbon cycle?

LESSON 1.5 · PRACTICE
Week 3 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.5: Photosynthesis

Name _____ Date _____

TODAY'S GOAL

I can explain how plants use light energy to make sugar and oxygen from carbon dioxide and water.

Standards: MS-LS1-6

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.5 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Analyze Data, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the answer for each item about photosynthesis.

1. Reactants of photosynthesis

2. Organelle where it happens

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · FLOATING LEAF DISKS

Does light intensity affect the rate of photosynthesis?

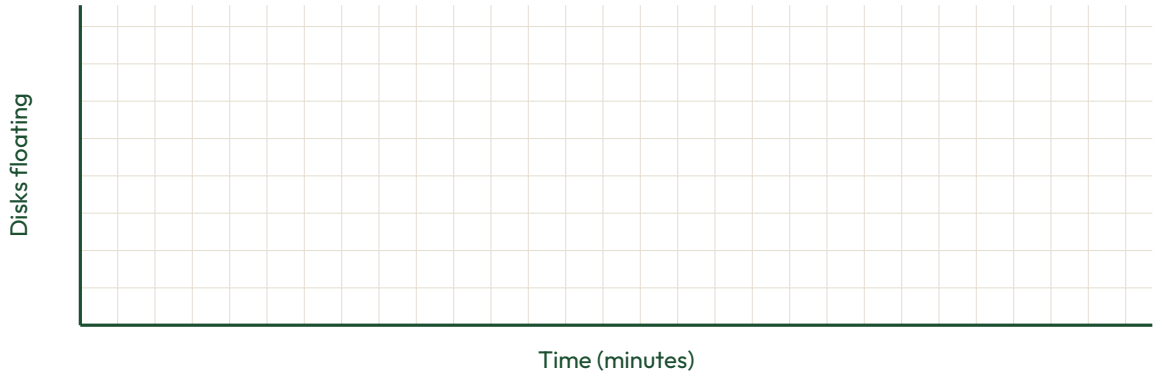
You need: fresh spinach leaves; a straw or hole punch; a plastic syringe (no needle); baking soda solution (1/8 tsp in 300 mL water) with 1 drop dish soap; 2 clear cups; a lamp.

1. Punch 20 disks from the leaves. ☐
2. Put 10 disks in the syringe with solution, push out the air, cover the tip, and pull the plunger to remove gas from the leaves until the disks sink. Repeat for the other 10. ☐
3. Put 10 sunk disks in a cup under the lamp and 10 in a cup in a dark place. ☐
4. Count floating disks every minute for 15 minutes. ☐

3 Record your results in the table.

Minute	Lamp cup: disks floating	Dark cup: disks floating
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		

4 Graph both cups on the same axes. Use a key.



5 Write a CER: Why did more disks float in one cup?

Claim My answer to the question	
Evidence Data from my investigation	
Reasoning Why my evidence supports my claim	

PART C Check Your Understanding

Directions: Answer each question.

6 Where does most of a tree's mass come from?

7 What are the products of photosynthesis?

8 What pigment captures light?

PART D Analyze Data

9 Disks floating after 10 min: lamp cup 9; dark cup 0. Which had more photosynthesis?

10 What gas made the disks float?

11 What was the independent variable?

12 Write a CER for the leaf disk results.

PART E Exit Ticket

13 Explain how photosynthesis transforms energy.

LESSON 1.6 · PRACTICE
Week 3 · Lesson 2 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.6: Cellular Respiration

Name _____ Date _____

TODAY'S GOAL

I can explain how cells release energy from food through cellular respiration.

Standards: MS-LS1-7

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.6 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Compare, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the answer for each item about cellular respiration.

1. Products of respiration

2. Reactants of respiration

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · YEAST BALLOONS

Does yeast need sugar to release carbon dioxide?

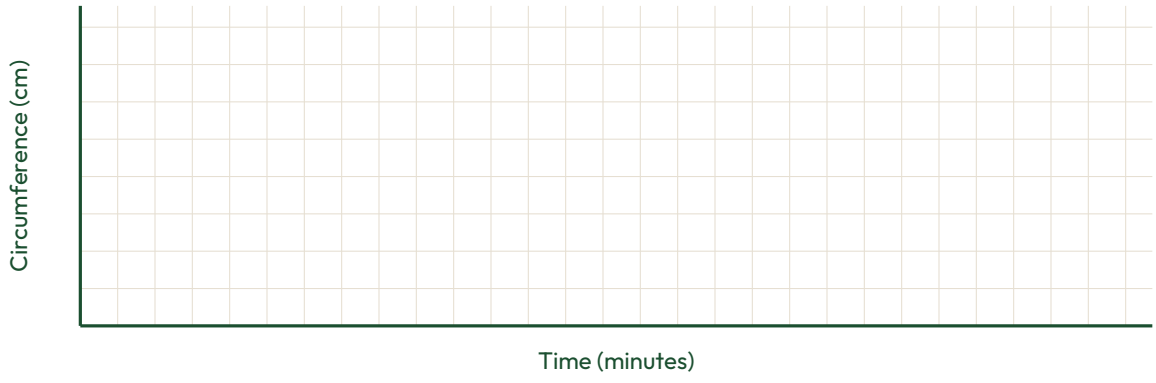
You need: 2 empty water bottles; 2 balloons; 2 packets of dry yeast; warm water; 2 tablespoons sugar.

1. Add 1 cup warm water and 1 packet yeast to each bottle. ☐
2. Add sugar to bottle A only. Swirl both. ☐
3. Stretch a balloon over each bottle's mouth. ☐
4. Measure each balloon's circumference every 10 minutes for 30 minutes. ☐

3 Record your results in the table.

Time	Balloon A, with sugar (cm)	Balloon B, no sugar (cm)
0 minutes		
10 minutes		
20 minutes		
30 minutes		

4 Graph both balloons. Use a key.



5 What filled the balloons? Why did one grow more?

PART C Check Your Understanding

Directions: Answer each question.

6 Where does cellular respiration happen?

7 What gas filled the balloon?

8 Do plants carry out respiration?

PART D Compare

Directions: Write photosynthesis or cellular respiration for each question.

9 Which process stores energy in glucose?

10 Which process releases energy from glucose?

11 Which produces oxygen?

12 Which produces carbon dioxide?

13 Balloon A (sugar) grew to 28 cm; balloon B (no sugar) to 13 cm. Explain using CER.

PART E Exit Ticket

14 Explain how matter cycles but energy flows between photosynthesis and respiration.

WEEKLY CHECK
After Lesson 1.6

About 15 minutes

Weekly Check 3: Photosynthesis and Respiration

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Write the answer for each item.

1. Products of photosynthesis

2. Products of cellular respiration

3. Organelle for photosynthesis

4. Organelle for respiration

5. Explain how photosynthesis and respiration cycle matter.

LESSON 1.7 · PRACTICE
Week 4 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.7: Food Webs and Energy Pyramids

Name _____ Date _____

TODAY'S GOAL

I can model how energy flows through a food web and explain why less energy reaches each higher level.

Standards: MS-LS2-3

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.7 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the term for each description.

1. An organism that makes its own food

2. Organisms that break down dead matter

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · BUILD A FOOD WEB

How are organisms in a forest ecosystem connected?

You need: cards: oak tree, grass, acorn weevil, grasshopper, mouse, rabbit, deer, frog, snake, hawk, fox, mushroom (decomposer); yarn or a pencil.

1. Arrange the cards on a large sheet of paper. ☐
2. Draw arrows from each food to the organism that eats it. ☐
3. Label producers, consumers, and the decomposer. ☐
4. Remove the mouse. Predict two effects on the web. ☐

3 Draw your food web. Arrows point from the food to the eater.

4 Predictions: what happens if one organism disappears?

PART C Check Your Understanding

Directions: Answer each question.

5 Which way do food-chain arrows point?

6 If grass has 20,000 units, about how much reaches grasshoppers?

7 Why are top predators few?

PART D Practice

8 Producers have 50,000 units. Primary consumers?

9 Secondary consumers?

10 Tertiary consumers?

11 In grass → rabbit → fox, the fox is a...

12 If a disease kills most rabbits, predict effects on the grass and the foxes. Explain.

PART E Exit Ticket

13 Explain the difference between how matter and energy move through an ecosystem.

LESSON 1.8 · PRACTICE
Week 4 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.8: Biodiversity and Ecosystem Services

Name _____ Date _____

TODAY'S GOAL

I can explain why biodiversity matters and evaluate ways to protect ecosystem services.

Standards: MS-LS2-5

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.8 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the answer for each item.

1. An animal that carries pollen

2. Wetlands help prevent...

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · BIODIVERSITY SURVEY

How many kinds of living things live in a small area?

You need: 4 sticks and string to mark a 1 m × 1 m square; magnifying glass; paper and a pencil.

1. Mark a square in a lawn and another in a less-mowed area (garden edge, field, or woods). ☐
2. Count the different kinds of plants and animals in each square for 10 minutes. ☐
3. Compare the two squares. ☐

3 Record your results in the table.

Square	Kinds of plants	Kinds of animals	Total kinds
Lawn			
Less-mowed area			

4 Which area has more biodiversity? Why?

PART C Check Your Understanding

Directions: Answer each question.

5 What is biodiversity?

6 Name two ecosystem services.

7 Why is a diverse ecosystem more stable?

PART D Practice

8 Lawn square: 3 species. Field square: 14 species. Which has more biodiversity?

9 Explain one way the loss of pollinators would affect people.

10 Evaluate: A farmer wants to remove all hedgerows to plant more crops. Name one benefit and one cost to ecosystem services.

PART E Exit Ticket

11 Choose one ecosystem service and explain how people could protect it.

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 1 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Practices and cycles

Directions: Write the term for each description.

1. The variable you measure

2. Plants releasing water vapor

3. Changing N_2 into usable forms

4. Burning fuel

PART B Energy

Directions: Answer each item. Show your math for the energy pyramid.

5. Photosynthesis equation (words)

6. Producers 30,000 units; primary consumers?

7. Why do energy pyramids narrow at the top?

UNIT TEST
At the end of the unit

About 45 minutes

Unit 1 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 20 points.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Science practices (1 point each)

Directions: Name each variable for the experiment.

1. Does the temperature of water affect how much sugar dissolves? Independent variable:

2. Dependent variable:

3. Name one controlled variable.

PART B Part B · Cycles (1 point each)

4. Which process adds water vapor to the air from plants?

A condensation
B transpiration
C infiltration
D runoff

5. What force pulls precipitation down?

6 Which process removes carbon dioxide from the atmosphere?

A combustion
B decomposition
C photosynthesis
D respiration

7 Why can't animals use nitrogen gas directly?

8 Name the organisms that return nutrients from dead matter to soil.

PART C Part C · Energy in living things (1 point each)

9 Write the word equation for photosynthesis.

10 Write the word equation for cellular respiration.

11 Producers store 40,000 units of energy. About how much reaches secondary consumers?

12 Which is an ecosystem service?

A a parking lot
B wetlands filtering water
C a concrete channel

PART D Part D · Data analysis (4 points)

Directions: Use the data to answer each part. Give a claim, evidence, and reasoning.

13 Leaf disks floating after 12 minutes: bright light 10, dim light 5, dark 0. Write a CER explaining the results.

PART E Part E · Explain (4 points)

14 Use a food chain (grass → rabbit → fox) to explain how matter cycles and energy flows. Include photosynthesis, respiration, and decomposers.

UNIT 2

Properties and Changes of Matter

Atoms and molecules, physical properties and states, changes of state, density, solubility and conductivity, chemical properties, and chemical changes.

Your practice pages for Unit 2. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

atom	The smallest particle of an element.
compound	Different elements chemically joined.
density	Mass per volume.
thermal energy	The energy of moving particles.
chemical change	A change that makes new substances.
conservation of mass	Mass is not created or destroyed in a reaction.

LESSON 2.1 · PRACTICE
Week 1 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.1: Atoms, Elements, and Molecules

Name _____ Date _____

TODAY'S GOAL

I can describe atoms, elements, molecules, and compounds and model simple molecules.

Standards: MS-PS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.1 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Formulas, Part E Classify, Part F Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. Anything with mass that takes up space

_____

2. Two or more atoms joined together

_____

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · MODEL MOLECULES

How can models show the structure of molecules?

You need: mini marshmallows and gumdrops (or clay in two or three colors); toothpicks.

1. Choose a color for each element: for example, white = hydrogen, red = oxygen, black = carbon. ☐
2. Build H_2O, O_2, and CO_2 using toothpicks as bonds. ☐
3. Build CH_4 (methane): one carbon in the center with four hydrogens. ☐
4. Sketch each model and write its formula. ☐

3 Record your results in the table.

Molecule	Formula	Sketch of my model

4 Color key (which color stands for which element):

PART C Check Your Understanding

Directions: Answer each question.

5 What is the difference between an element and a compound?

6 How many atoms are in one molecule of CO_2?

7 Is O_2 an element or a compound?

PART D Formulas

Directions: For each formula, write the total number of atoms and how many of each element.

8 H_2O_2 (hydrogen peroxide)

9 NaCl (table salt)

10 $C_6H_{12}O_6$ (glucose)

11 He (helium)

PART E Classify

Directions: Write element or compound for each substance.

12 N_2

13 CO

14 Fe

15 CH_4

PART F Exit Ticket

Directions: Answer in complete sentences.

16 Why does water have different properties from hydrogen and oxygen?

LESSON 2.2 · PRACTICE
Week 1 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.2: Physical Properties and States of Matter

Name _____ Date _____

TODAY'S GOAL

I can describe physical properties and compare solids, liquids, and gases.

Standards: MS-PS1-2, MS-PS1-4

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.2 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. The amount of matter in an object

_____

2. The amount of space something takes up

_____

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · PROPERTY DETECTIVE

Which properties can identify an unknown substance?

You need: small samples of salt, sugar, baking soda, and cornstarch; a magnifying glass; cups of water; vinegar; a data table.

1. Observe each powder with the magnifying glass and describe its crystals. ☐
2. Rub a pinch between your fingers and describe its texture. ☐
3. Stir a spoonful into water: does it dissolve? ☐
4. Have someone give you a mystery sample and identify it from your table. ☐

Safety: Don't taste any samples. Wear safety goggles.

3 Record your results in the table.

Powder	Crystals (magnifier)	Texture	Dissolves in water?

4 The mystery sample is _____. My evidence:

5 Which properties from your Property Detective table were most useful for identifying the mystery sample? Explain.

PART C Check Your Understanding

Directions: Answer each question.

6 Name three physical properties.

7 Which state has a definite volume but no definite shape?

8 How do gas particles move?

PART D Practice

Directions: Write solid, liquid, or gas for each description.

9 Particles vibrate in place

10 Takes the shape of its container but keeps its volume

11 Fills any container

12 Particles slide past each other

PART E Exit Ticket

Directions: Answer in complete sentences.

13 Is boiling point a physical property? Explain.

WEEKLY CHECK
After Lesson 2.2

About 15 minutes

Weekly Check 1: Atoms and Properties

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question about atoms and properties.

1. A pure substance made of one kind of atom

2. Number of atoms in H_2O

3. Element or compound: CO_2

4. State with definite volume but no definite shape

5. Physical property: name one

6. Describe how particles move in a solid and in a gas.

LESSON 2.3 · PRACTICE
Week 2 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.3: Particles in Motion: Changes of State

Name _____ Date _____

TODAY'S GOAL

I can explain how adding or removing thermal energy changes particle motion, temperature, and state.

Standards: MS-PS1-4

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.3 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. Solid directly to gas

2. Liquid to gas at the surface

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · ICE MELT DATA

What happens to temperature while ice melts?

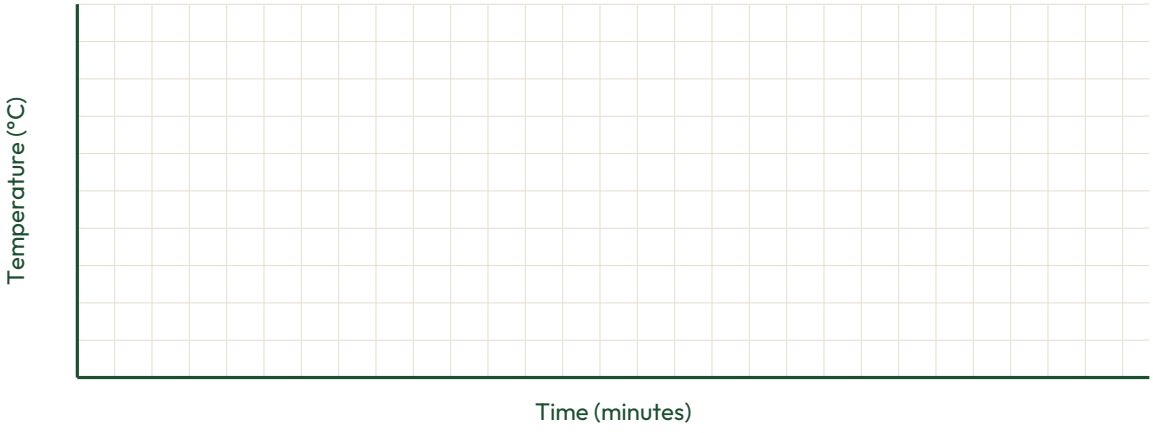
You need: a cup of crushed ice; a thermometer; a timer; a stirring spoon.

1. Measure the starting temperature of the ice. ☐
2. Let it sit at room temperature; stir and record the temperature every 2 minutes. ☐
3. Note the time when the ice is half melted and fully melted. ☐
4. Keep recording for 10 more minutes. ☐

3 Record the temperature every 2 minutes. Note when the ice is half melted and fully melted.

Time (min)	Temp (°C)	Time (min)	Temp (°C)

4 Graph temperature over time. Label the part where the ice was melting.



5 Describe your Ice Melt graph. What happened to the temperature while the ice melted, and why?

PART C Check Your Understanding

Directions: Answer each question.

6 What happens to particle motion when thermal energy is added?

7 Name the change from gas to liquid.

8 Why does the temperature stay at 0°C while ice melts?

PART D Practice

Directions: Name each change of state and tell whether energy is added or removed.

9 a mirror fogging after a shower

10 butter softening in a pan

11 a lake freezing

12 frost disappearing on a cold, sunny morning without melting

PART E Exit Ticket

Directions: Answer in complete sentences.

13 Explain, in terms of particles, why a puddle evaporates on a sunny day.

LESSON 2.4 · PRACTICE
Week 2 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.4: Density

Name _____ Date _____

TODAY'S GOAL

I can calculate density and use it to predict whether an object will sink or float.

Standards: MS-PS1-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.4 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Calculate, Part E Apply, Part F Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. Mass packed into a volume

2. The upward push of a liquid

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · DENSITY COLUMN

Can liquids of different densities stack?

You need: a tall clear glass; honey, dish soap, water with food coloring, vegetable oil; small objects: a grape, a cork, a plastic bead.

1. Slowly pour about 2 cm of each liquid, in this order: honey, dish soap, colored water, oil. ☐
2. Watch whether the layers mix. ☐
3. Gently drop in each small object. ☐
4. Record where each object stops. ☐

3 Draw your column. Label each layer from densest (bottom) to least dense (top), and show where each object stopped.

PART C Check Your Understanding

Directions: Answer each question.

4 A rock has a mass of 30 g and a volume of 10 cm^3. What is its density?

5 Will an object with a density of 0.6 g/cm³ float in water?

6 How can you find the volume of an irregular object?

PART D Calculate

Directions: Find each density. Show your work. Use the table to name the substance when you can.

7 m = 40 g, V = 20 cm³

8 m = 79 g, V = 10 cm³

9 m = 12 g, V = 50 cm³

10 m = 193 g, V = 10 cm³

PART E Apply

Directions: Show your work.

11 A water level rises from 50 mL to 65 mL when a stone is dropped in. The stone's mass is 45 g. Find its volume and density, and predict whether it floats.

PART F Exit Ticket

Directions: Answer in complete sentences.

12 Explain why a steel ship floats but a steel nail sinks.

WEEKLY CHECK
After Lesson 2.4

About 15 minutes

Weekly Check 2: State Changes and Density

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question. Show your work for calculations.

1. Gas to liquid

2. Solid directly to gas

3. Density formula

4. Density of a 60 g, 20 cm³ block

5. Will a 0.8 g/cm³ object float in water?

6. Why does temperature stay the same while ice melts?

LESSON 2.5 · PRACTICE
Week 3 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.5: Solubility, Conductivity, and Mixtures

Name _____ Date _____

TODAY'S GOAL

I can describe solubility and conductivity and tell mixtures and solutions apart.

Standards: MS-PS1-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.5 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. The substance that dissolves

2. How well a material passes heat or electricity

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · SEPARATE A MIXTURE

Can you separate sand, salt, and iron filings using their properties?

You need: a mixture of sand, salt, and iron filings (or paper clips); a magnet in a plastic bag; a coffee filter and funnel; water; a shallow dish.

1. Pass the magnet over the mixture to remove the iron. ☐
2. Stir the rest into warm water so the salt dissolves. ☐
3. Pour it through the filter to catch the sand. ☐
4. Leave the salty water in the dish for a few days to evaporate. ☐

Safety: Wear goggles. Keep the iron filings in a bag on the magnet.

3 Draw a flowchart of your steps. Name the property you used at each step.

4 Explain which property you used at each step of Separate a Mixture.

PART C Check Your Understanding

Directions: Answer each question.

5. Is salt water a mixture or a compound?

6. How could you separate iron filings from sand?

7. Does sugar usually dissolve better in hot or cold water?

PART D Practice

Directions: Tell whether each is a solution or another mixture, and name one way to separate it.

8. lemonade with sugar dissolved

9. pebbles in water

10. paper clips in rice

11. muddy water

PART E Exit Ticket

Directions: Answer in complete sentences.

12 Why are electric wires made of copper and coated in plastic?

LESSON 2.6 · PRACTICE
Week 3 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.6: Chemical Properties

Name _____ Date _____

TODAY'S GOAL

I can describe chemical properties and explain how they differ from physical properties.

Standards: MS-PS1-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.6 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Classify, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. How easily a substance burns

2. A reaction with oxygen

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · RUST RACE

Which conditions make iron rust fastest?

You need: four steel wool pads or iron nails; four cups; water, salt water, vinegar, and an empty (dry) cup; labels.

1. Place one pad or nail in each cup: dry, water, salt water, vinegar. ☐
2. Label the cups and predict which will rust fastest. ☐
3. Observe each day for five days. ☐
4. Record color, texture, and amount of rust. ☐

Safety: Wear goggles and gloves. Wash hands after handling steel wool.

3 Prediction: which cup will rust fastest? Why?

4 Each day, record color, texture, and amount of rust.

Day	Dry	Water	Salt water	Vinegar
1				
2				
3				
4				
5				

5 Which Rust Race cup rusted fastest? Use your observations as evidence.

PART C Check Your Understanding

Directions: Answer each question.

6 Physical or chemical property: iron rusts

7 Physical or chemical property: ice melts at 0°C

8 Why is gold used in electronics?

PART D Classify

Directions: Write physical property or chemical property for each description.

9 silver tarnishes black

10 aluminum is light and silvery

11 gasoline catches fire easily

12 sugar dissolves in water

13 baking soda fizzes in vinegar

14 water boils at 100°C

PART E Exit Ticket

Directions: Answer in complete sentences.

15 Explain why you can't observe a chemical property without changing the substance.

WEEKLY CHECK
After Lesson 2.6

About 15 minutes

Weekly Check 3: Mixtures and Chemical Properties

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question about mixtures and properties.

1. A mixture so evenly blended it looks like one substance

2. Separate iron filings from sand with a...

3. Physical or chemical property: iron rusts

4. Physical or chemical property: copper conducts electricity

5. How easily a substance burns

6. Explain the difference between a physical and a chemical property.

LESSON 2.7 · PRACTICE
Week 4 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.7: Physical and Chemical Changes

Name _____ Date _____

TODAY'S GOAL

I can tell physical changes from chemical changes using evidence.

Standards: MS-PS1-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.7 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Check Your Understanding, Part C Classify, Part D Explain, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. A new substance made by a reaction

2. A solid formed when two liquids react

PART B Check Your Understanding

Directions: Answer each question.

3. Physical or chemical change: melting butter

4. Physical or chemical change: burning a match

5 Name two kinds of evidence of a chemical change.

PART C Classify

Directions: Write physical or chemical change for each example.

6 chopping wood

7 frying an egg

8 dissolving sugar in water

9 a nail rusting

10 water freezing into ice

11 a cake baking

PART D Explain

Directions: Answer in complete sentences.

12 Boiling water forms bubbles. Is boiling a chemical change? Explain using evidence and the definition.

PART E Exit Ticket

Directions: Answer in complete sentences.

13 Prove that toasting bread is a chemical change. Name two pieces of evidence.

LESSON 2.8 · PRACTICE
Week 4 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.8: Lab: Evidence of a Chemical Reaction

Name _____ Date _____

TODAY'S GOAL

I can plan and carry out a lab, collect evidence of a chemical reaction, and write a claim, evidence, and reasoning.

Standards: MS-PS1-2, MS-PS1-5

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.8 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. The factor you measure

2. A testable prediction with a reason

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · BALLOON IN A BOTTLE

Is mass conserved in a chemical reaction?

You need: a kitchen scale; a small plastic bottle; a balloon; baking soda (1 teaspoon); vinegar (about 50 mL); a funnel.

1. Pour the vinegar into the bottle. Use the funnel to put the baking soda into the balloon. ☐
2. Stretch the balloon over the bottle opening without spilling the baking soda. Measure the total mass. ☐
3. Lift the balloon so the baking soda falls in. Record your observations. ☐
4. When the fizzing stops, measure the total mass again. Then repeat with the bottle open (no balloon) and compare. ☐

Safety: Wear goggles. Point the bottle away from faces.

3 Write your hypothesis for the sealed-bottle trial in If… then… because… form.

4 Record what you see during each trial.

Trial	What I observed
Sealed (balloon)	
Open (no balloon)	

5 sealed: mass before (g)

6 sealed: mass after (g)

7 open: mass before (g)

8 open: mass after (g)

9 List two pieces of evidence that a chemical reaction happened.

10 Write a CER paragraph answering the lab question.

PART C Check Your Understanding

Directions: Answer each question.

11 What gas forms when baking soda reacts with vinegar?

12 Why does the open container lose mass?

13 What does conservation of mass mean?

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 2 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Vocabulary

Directions: Write what each word means.

1. compound

2. density

3. sublimation

4. solution

5. chemical property

6. precipitate

PART B Practice

Directions: Answer each question. Show your work.

7. Density of 45 g in 15 cm^3

8. Atoms in C_2H_6

9. Name three pieces of evidence that a chemical change has happened.

UNIT TEST
At the end of the unit

About 45 minutes

Unit 2 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 16 points.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Atoms and molecules (1 point each)

Directions: Answer each question. For questions with choices, circle the letter of the best answer.

1. Which is a compound?

   A O_2
   B Fe
   C H_2O
   D He

2. How many atoms are in one molecule of CH_4?

3. What is the difference between an atom and a molecule?

PART B Part B · Properties and states (1 point each)

Directions: Answer each question. For questions with choices, circle the letter of the best answer. Show your work for calculations.

4. Name the change from liquid to solid.

5 When thermal energy is added to a substance, its particles…

A slow down
B speed up
C stop moving
D get bigger

6 Find the density: m = 24 g, V = 8 cm³.

7 Will the object in the last question sink or float in water?

8 Which is a chemical property?

A boiling point
B color
C flammability
D density

PART C Part C · Mixtures and changes (1 point each)

Directions: Answer each question. For questions with choices, circle the letter of the best answer.

9 How could you separate salt from salt water?

10 Which is a chemical change?

A ice melting
B paper being cut
C wood burning
D sugar dissolving

11 Name two kinds of evidence of a chemical change.

12 In a sealed container, what happens to total mass during a reaction?

PART D Part D · Explain (4 points)

Directions: Write a paragraph. Use science words from this unit.

13 A student burns a candle on a scale in an open room and finds the mass goes down. Another student says this proves matter was destroyed. Explain whether they are right, using what you know about chemical changes and conservation of mass.

Grade 7 Science

Student Workbook

A full year of Grade 7 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Workbook holds the daily practice pages for every lesson, plus a Weekly Check at the end of each week, the unit reviews, and the unit tests, with room to show thinking and write complete answers.

INSIDE THIS YEAR

01 Cycles of Matter and Energy

02 Properties and Changes of Matter

IN THIS BOOK

- Warm-ups, practice, and exit tickets for every lesson
- A Weekly Check for each week of lessons
- Unit reviews and unit tests
- Space to show work and explain thinking

THE COMPLETE SET



Textbook

Workbook

You are holding this book

Teacher Edition

Each lesson has the same number in all three books, so the textbook, workbook, and teacher pages always match.